CLASSWORK 11

Logistics Regression by Hand

CPE 486/586

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20 points

A maintenance team monitors cooling fans in a server room. For a fan, they record two integer readings on a four-level scale with values 0, 1, 2 and 3. Here x_1 is the vibration level and x_2 is the lubrication level. The label is $y = 1$ when the fan failed within 30 days and $y = 0$ when it kept running. Table 1 lists the five fans used for training.

Table 1. Training data, $N = 5$.

Fan i	x_{i1} (vibration)	x_{i2} (lubrication)	y_i
1	3	0	1
2	1	2	0
3	3	1	1
4	0	1	0
5	2	2	0

A new technician has some initial weight to start with. You'll start from these weights and work on determining how gradient descent evolve them.

$$\boldsymbol{\theta}^{(0)} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \\ 0.5 \end{bmatrix}, \qquad \eta = 1.0, \qquad N = 5.$$

Equations to use

Augmented input:	$\mathbf{x}^{(i)} = \begin{bmatrix} 1 & x_{i1} & x_{i2} \end{bmatrix}^\top$, the leading 1 for the bias θ_0
Linear score:	$z_i = \boldsymbol{\theta}^\top \mathbf{x}^{(i)} = \theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2}$
Hypothesis:	$h_i = h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) = g(z_i) = \dfrac{1}{1+e^{-z_i}}$
Cost (natural log):	$J(\boldsymbol{\theta}) = -\dfrac{1}{N}\sum_{i=1}^{N}\Big[y_i \ln h_i + (1-y_i)\ln(1-h_i)\Big] = \dfrac{1}{N}\sum_{i=1}^{N}\ell_i$
Per-sample loss:	$\ell_i = -\ln h_i$ when $y_i = 1$, $\quad \ell_i = -\ln(1-h_i)$ when $y_i = 0$
Error:	$e_i = h_i - y_i$
Gradient:	$\dfrac{\partial J}{\partial \theta_j} = \dfrac{1}{N}\sum_{i=1}^{N} e_i\, x_{ij} = \dfrac{S_j}{N}$, $\quad$ with $x_{i0} = 1$
Update:	$\theta_j^{(t+1)} = \theta_j^{(t)} - \eta \left.\dfrac{\partial J}{\partial \theta_j}\right\vert_{\boldsymbol{\theta}^{(t)}}$ $\quad$ for $j = 0, 1, 2$
Prediction:	$\hat{y} = 1$ when $h \geq 0.5$, which happens exactly when $z \geq 0$
Decision boundary:	$\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$, $\quad$ so $\quad x_2 = -\dfrac{\theta_0 + \theta_1 x_1}{\theta_2}$

Sigmoid Properties

(P1)	$g(0) = \frac{1}{2}$	A score of zero at the boundary.
(P2)	$g(-z) = 1 - g(z)$	
(P3)	$\ln\big(1 - g(z)\big) = \ln g(-z)$	For $z < 0$, the two loss columns of Table 2 swap.
(P4)	$g'(z) = g(z)\big(1 - g(z)\big)$	With the chain rule, it gives $\partial \ell_i / \partial z_i = h_i - y_i$.

A note on signs. The slides define the log-likelihood $LL(\boldsymbol{\theta}) = \sum_i \big[y_i \ln h_i + (1-y_i)\ln(1-h_i)\big]$, so $LL = -NJ$. Maximizing LL and minimizing J lead to the same $\boldsymbol{\theta}$. Here you minimize J, so the update subtracts the gradient of J. In the form of the slides, $\theta_j \leftarrow \theta_j + \frac{\eta}{N}\sum_i (y_i - h_i)\, x_{ij}$, which is the same step.

Rules for Hand Computation

R1. Compute z_i with max three decimal places.

R2. Round z_i to the nearest 0.1 only to read Table 2. Round a halfway value away from zero.

R3. Copy h_i and ℓ_i from Table 2 with four decimals. Keep four decimals for e_i, $e_i x_{ij}$, S_j and $\partial J / \partial \theta_j$.

R4. Round $\theta_j^{(t+1)}$ to three decimals before the next iteration.

R5. Report J with four decimals.

Steps

1. Compute $z_i = \theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2}$ with the current $\boldsymbol{\theta}^{(t)}$.
2. Read $h_i = g(z_i)$ from Table 2.
3. Read ℓ_i from Table 2, in the column that matches y_i.
4. Compute $e_i = h_i - y_i$.
5. Compute $e_i x_{i1}$ and $e_i x_{i2}$.
6. Add down the columns. $S_0 = \sum_i e_i$, $S_1 = \sum_i e_i x_{i1}$, $S_2 = \sum_i e_i x_{i2}$.
7. Set $\partial J/\partial \theta_j = S_j/5$ and $\theta_j^{(t+1)} = \theta_j^{(t)} - \eta\, \partial J/\partial \theta_j$. Update $j = 0, 1, 2$ from the old $\boldsymbol{\theta}^{(t)}$ together.
8. Compute $J(\boldsymbol{\theta}^{(t)}) = \frac{1}{5}\sum_i \ell_i$ and count the fans with $\hat{y}_i = y_i$. These describe the weights before the update.

Table 2. Sigmoid values and per-sample losses for $z \geq 0$. Apply (P2) and (P3) for $z < 0$.

z	$g(z)$	$-\ln g(z)$ loss if $y=1$	$-\ln(1-g(z))$ loss if $y=0$
0.0	0.5000	0.6931	0.6931
0.1	0.5250	0.6444	0.7444
0.2	0.5498	0.5981	0.7981
0.3	0.5744	0.5544	0.8544
0.4	0.5987	0.5130	0.9130
0.5	0.6225	0.4741	0.9741
0.6	0.6457	0.4375	1.0375
0.7	0.6682	0.4032	1.1032
0.8	0.6900	0.3711	1.1711
0.9	0.7109	0.3412	1.2412
1.0	0.7311	0.3133	1.3133
1.1	0.7503	0.2873	1.3873
1.2	0.7685	0.2633	1.4633
1.3	0.7858	0.2410	1.5410
1.4	0.8022	0.2204	1.6204
1.5	0.8176	0.2014	1.7014

z	$g(z)$	$-\ln g(z)$ loss if $y=1$	$-\ln(1-g(z))$ loss if $y=0$
1.6	0.8320	0.1839	1.7839
1.7	0.8455	0.1678	1.8678
1.8	0.8581	0.1530	1.9530
1.9	0.8699	0.1394	2.0394
2.0	0.8808	0.1269	2.1269
2.1	0.8909	0.1155	2.2155
2.2	0.9002	0.1051	2.3051
2.3	0.9089	0.0955	2.3955
2.4	0.9168	0.0868	2.4868
2.5	0.9241	0.0789	2.5789
2.6	0.9309	0.0716	2.6716
2.7	0.9370	0.0650	2.7650
2.8	0.9427	0.0590	2.8590
2.9	0.9478	0.0536	2.9536
3.0	0.9526	0.0486	3.0486

Example of a negative lookup. Take $z = -0.2$. Property (P2) gives $g(-0.2) = 1 - 0.5498 = 0.4502$. For $y = 1$, the loss is $-\ln 0.4502$, read from the $y = 0$ column at 0.2, so $\ell = 0.7981$. For $y = 0$, the loss comes from the $y = 1$ column at 0.2, so $\ell = 0.5981$.

Part 1: Iteration 1, $\boldsymbol{\theta}^{(0)} \to \boldsymbol{\theta}^{(1)}$ (5 points)

Start from $\boldsymbol{\theta}^{(0)} = (\theta_0, \theta_1, \theta_2) = (0.5,\ -0.5,\ 0.5)$.

(a) **[3 points]** Complete Table 3 with the first six steps.

Table 3. Forward pass and errors at $\boldsymbol{\theta}^{(0)}$.

Fan i	x_{i1}	x_{i2}	y_i	z_i	h_i	ℓ_i	$e_i = h_i - y_i$	$e_i\, x_{i1}$	$e_i\, x_{i2}$
1	3	0	1	-1.0	0.2689	1.3133	-0.7311	-2.1933	0
2	1	2	0						
3	3	1	1						
4	0	1	0						
5	2	2	0						
					Column sums		$S_0 =$	$S_1 =$	$S_2 =$

(b) **[2 points]** Complete Table 4 with step 7, then finish step 8.

Table 4. Gradient and update, $\eta = 1.0$.

j	$\theta_j^{(0)}$	S_j (Table 3)	$\frac{\partial J}{\partial \theta_j} = \frac{S_j}{5}$	$\theta_j^{(1)} = \theta_j^{(0)} - \eta \frac{\partial J}{\partial \theta_j}$
0	0.5	0.7311	0.1462	0.354
1	−0.5			
2	0.5			

$J(\boldsymbol{\theta}^{(0)}) = \frac{1}{5}\sum_i \ell_i =$ _____ Fans with $\hat{y}_i = y_i$ under $\boldsymbol{\theta}^{(0)}$: _____ of 5

Part 2: Iteration 2, $\boldsymbol{\theta}^{(1)} \to \boldsymbol{\theta}^{(2)}$ (5 points)

Copy $\boldsymbol{\theta}^{(1)}$ from Table 4. $\theta_0 =$ _____ $\theta_1 =$ _____ $\theta_2 =$ _____

(a) **[3 points]** Complete Table 5 with the first six steps.

Table 5. Forward pass and errors at $\boldsymbol{\theta}^{(1)}$.

Fan i	x_{i1}	x_{i2}	y_i	z_i	h_i	ℓ_i	$e_i = h_i - y_i$	$e_i\, x_{i1}$	$e_i\, x_{i2}$
1	3	0	1	$0.105 \to 0.1$	0.5250	0.6444	−0.4750	−1.4250	0
2	1	2	0						
3	3	1	1						
4	0	1	0						
5	2	2	0						
					Column sums		$S_0 =$	$S_1 =$	$S_2 =$

(b) **[2 points]** Complete Table 6 with step 7, then finish step 8.

Table 6. Gradient and update, $\eta = 1.0$.

j	$\theta_j^{(1)}$	S_j (Table 5)	$\frac{\partial J}{\partial \theta_j} = \frac{S_j}{5}$	$\theta_j^{(2)} = \theta_j^{(1)} - \eta \frac{\partial J}{\partial \theta_j}$
0	0.354	0.6494	0.1299	0.224
1				
2				

$J(\boldsymbol{\theta}^{(1)}) = \frac{1}{5}\sum_i \ell_i =$ _____ Fans with $\hat{y}_i = y_i$ under $\boldsymbol{\theta}^{(1)}$: _____ of 5

Part 3: Iteration 3, $\boldsymbol{\theta}^{(2)} \to \boldsymbol{\theta}^{(3)}$ (5 points)

Copy $\boldsymbol{\theta}^{(2)}$ from Table 6. $\theta_0 =$ _____ $\theta_1 =$ _____ $\theta_2 =$ _____

(a) **[3 points]** Complete Table 7 with the first six steps.

Table 7. Forward pass and errors at $\boldsymbol{\theta}^{(2)}$.

Fan i	x_{i1}	x_{i2}	y_i	z_i	h_i	ℓ_i	$e_i = h_i - y_i$	$e_i\, x_{i1}$	$e_i\, x_{i2}$
1	3	0	1	$0.785 \to 0.8$	0.6900	0.3711	-0.3100	-0.9300	0
2	1	2	0						
3	3	1	1						
4	0	1	0						
5	2	2	0						
					Column sums		$S_0 =$	$S_1 =$	$S_2 =$

(b) **[2 points]** Complete Table 8 with step 7, then finish step 8.

Table 8. Gradient and update, $\eta = 1.0$.

j	$\theta_j^{(2)}$	S_j (Table 7)	$\frac{\partial J}{\partial \theta_j} = \frac{S_j}{5}$	$\theta_j^{(3)} = \theta_j^{(2)} - \eta \frac{\partial J}{\partial \theta_j}$
0	0.224	0.4456	0.0891	0.135
1				
2				

$J(\boldsymbol{\theta}^{(2)}) = \frac{1}{5}\sum_i \ell_i =$ _____ Fans with $\hat{y}_i = y_i$ under $\boldsymbol{\theta}^{(2)}$: _____ of 5

Part 4: The Trained Model (5 points)

(a) **[2 points]** Evaluate $\boldsymbol{\theta}^{(3)}$ on the training data with the first three steps, then compute $J(\boldsymbol{\theta}^{(3)})$ and fill the prediction columns.

Table 9. Training data under $\boldsymbol{\theta}^{(3)}$.

Fan i	x_{i1}	x_{i2}	y_i	z_i	h_i	ℓ_i	$\hat{y}_i$	Correct?
1	3	0	1	$1.326 \to 1.3$	0.7858	0.2410	1	yes
2	1	2	0					
3	3	1	1					
4	0	1	0					
5	2	2	0					
					Column sum			

$$J(\boldsymbol{\theta}^{(3)}) = \tfrac{1}{5}\textstyle\sum_i \ell_i = _____$$

(b) **[1 point]** Copy your results into Table 10.

Table 10. Training history.

t	$\theta_0^{(t)}$	$\theta_1^{(t)}$	$\theta_2^{(t)}$	$J(\boldsymbol{\theta}^{(t)})$	Fans correct (of 5)
0	0.5	−0.5	0.5	1.1776	0
1	0.354	−0.083	−0.063	0.7361	2
2					
3					

(c) **[1 point]** A new fan reads $\mathbf{x}_{\text{new}} = (x_1, x_2) = (2, 1)$. Apply $\boldsymbol{\theta}^{(3)}$.

$z_{\text{new}} = \theta_0 + 2\theta_1 + \theta_2 =$ _____

$h_{\text{new}} = g(z_{\text{new}}) =$ _____

Predicted class $\hat{y}_{\text{new}}$: _____

(d) **[1 point]** The dashed line in Figure 1 is the boundary of $\boldsymbol{\theta}^{(0)}$, namely $x_2 = x_1 - 1$. Compute two points on the boundary of $\boldsymbol{\theta}^{(3)}$ with two decimals, then draw the line through them.

At $x_1 = 0$: $x_2 =$ _____ At $x_1 = 3$: $x_2 =$ _____

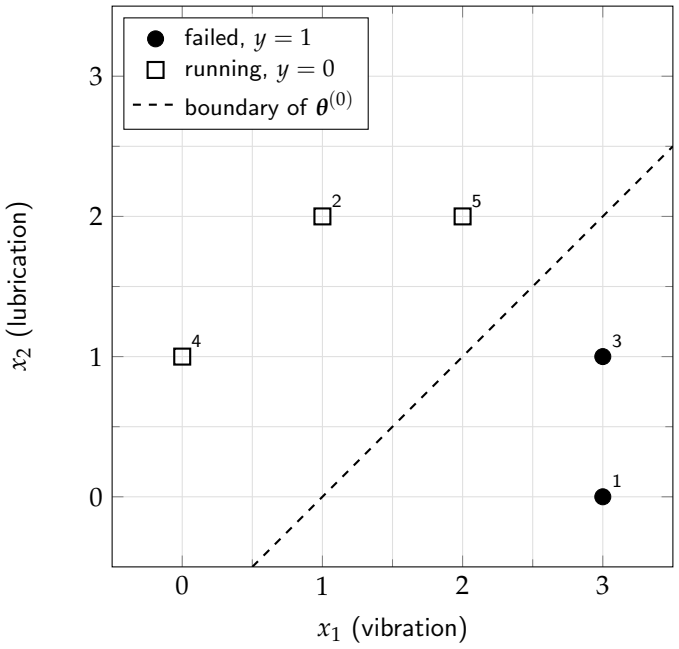


Figure 1. The five training fans, labeled by i. Draw the boundary of $\boldsymbol{\theta}^{(3)}$ here.